1) Thresholding and Intensity Level slicing PCB inspection

Given

$r = [25, 60, 95, 130, 165, 200, 237)$

Solder Joint intensity band = 90-170

(a) Binary threshold transformation

Given,

$$s = \begin{cases} 255, & r \geq 128 \\ 0, & r < 128 \end{cases}$$

r	25	60	95	130	165	200	235
s	0	0	0	255	255	255	255

output :-

$[0, 0, 0, 255, 255, 255, 255)$

b) intensity level slicing.

Pixels between 90 and 170 are changed to 255.

others remain unchanged.

* 20 → 25
* 60 → 60
* 95 → 255
* 130 → 255
* 165 → 255
* 200 → 200
* 235 → 235

Therefore $[25, 60, 255, 255, 255, 200, 235]$

(c) Number and percentage of highlighted pixels

Highlighted pixels are:

95, 130, 165

Number:

3 pixels

Total pixels = 7

$$\text{percentage} = \frac{3}{7} \times 100$$

$$= \boxed{42.86}$$

(d) Mean intensity

original mean:

$$\bar{r} = \frac{25+60+95+130+165+200+235}{7}$$

$$= \frac{910}{7}$$

$$\bar{r} = 130$$

slice output mean:

$$\bar{s} = \frac{25+60+255+255+255+200+235}{7}$$

$$= \frac{1285}{7}$$

$$\bar{s} = 183.57$$

e) Appropriate transformation.

Intensity level slicing is more appropriate for isolati the solder joints because the required solder - Joi range is specifically 90 - 170

Piecewise- Linear contrast stretching - X Ray image Enhancement

Control points : $(r_1, s_1) = (70, 20)$

$(r_2, s_2) = (140, 230)$

$L = 256$

$r = [30, 70, 90, 110, 140, 200]$

(a) Equations and slopes

Segment 1 : 0 - 70

Points :

$(0,0), (70, 20)$

Slope:

$$m_1 = \frac{20-0}{70-0} = \frac{20}{70} = \boxed{0.2857}$$

Equation: $s = 0.2857r$

Segment 2 : 70 - 140

$$m_2 = \frac{230-20}{140-70} = \frac{210}{70} = 3$$

using point slope form : $s - 20 = 3(r - 70)$

$$\boxed{s = 3r - 190}$$

Segment 3 : 140 - 255

Points :

$(140, 230), (255, 255)$

$$m_3 = \frac{255-230}{255-140} = \frac{25}{115}$$

$$\boxed{m_3 = 0.2174}$$

b) Transformed values

input r	segment	outputs
30	0-70	8.57
70	70-140	20
90	70-140	80
110	70-140	140
140	70-140	230
200	140-255	243.04

$S = [8.57, 20, 80, 140, 230, 243.04]$

(c) width and contrast gain

original diagnostic band : 140 - 70 = 70

So, original width : 70

After transformation : 230 - 20 = 210

Transformed width = 210

contrast gain :

$$G = \frac{210}{70}$$

$G = 3$

(d) compression and expansion

* **segment 1** slope = 0.2857 < 1 → intensity range **compressed**

* **segment 2** slope = 3 > 1 → intensity range is **Expanded**

segment 3 slope = 0.2174 < 1 → intensity range is **compressed**

c) Interpretation

The transformation improves visibility of the diagnostic region 70-140 because its intensity range increases from 70 levels to 210 levels.

3) Spatial filtering - Impulse noise Removal

The 3x3 neighbourhood is

$$\begin{bmatrix} 52 & 55 & 58 \\ 54 & \boxed{255} & 57 \\ 53 & 56 & 59 \end{bmatrix}$$

255 = Salt noise

(a) 3x3 mean filter

$$\text{Mean} = \frac{52*55 + 58 + 54 + 255 + 57 + 53 + 56 + 59}{9}$$

$$= \frac{699}{9}$$

Mean = 77.67

(b) weighted average filter :-

Mask :

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Weighted sum:

$$\frac{1}{16}[52+2(55)+58+2(54)+4(255)+2(57)+58+2(56)+59]$$

$$= \frac{1686}{16}$$

$$= \boxed{105.375}$$

(c) Median filter:

Average values in ascending order

52, 53, 54, 55, 56, 57, 58, 59, 255

The middle value is

56

$$\boxed{\text{Median filter} = 56}$$

(d) Absolute error

$$R = \frac{52+55+58+54+57+53+56+59}{8}$$

$$= \frac{444}{8}$$

$$\boxed{R = 55.5}$$

Mean filter error :-

$$\begin{bmatrix} 77.67 & -55.5 \\ 22.17 & \end{bmatrix}$$

weighted average filter error:

$$105.375 - 55.5$$

$$\boxed{49.875}$$

Median filter error

$$56 - 55.5$$

$$\boxed{0.5}$$

filter	output	Absolute error
Mean	77.67	22.17
weighted Avg	105.375	49.875
Median	56	0.5

(e) Best filter

The **median filter** is the **most appropriate**

⇒ It removes extreme salt noise value 255 effectively

⇒ smallest error = 0.5

4) Edge Detection – Sobel and Prewitt Gradient Analysis

Given region

$$\begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 90 \\ 10 & 90 & 90 \end{bmatrix}$$

(a) Sobel Gx and Gy

Sobel Gx : $Gx = (-1)(10) + 0(10) + (1)(10) + (-2)(10) + 0(10) + (2)(90) + (-1)(10) + 0(90) + 1(90)$

$$\boxed{Gx = 240}$$

sobel Gy : $Gy = (-1)((0) + (-2)(10) + (-1)(10) + 0 + 0 + 0 + (1)(10) + (2)(90) + 1(90)$

$$\boxed{Gy = 240}$$

(b) Gradient magnitude

Exact magnitude :-

$$M = \sqrt{G_x^2 + G_y^2}$$

$$= \sqrt{240^2 + 240^2}$$

$$\boxed{= 339.41}$$

Approximation :-

$$M = |Gx| + |Gy|$$

$$= 240 + 240$$

$$\boxed{= 480}$$

Percentage error :-

$$\frac{480 - 339.41}{339.41} \times 100$$

$$\boxed{= 41.42\%}$$

(e) Edge detection · for $M \geq 250$

Sobel

$$M_{sobel} = 339.41$$

Since

$$339.41 \geq 250$$

Sobel detects an edge

Prewitt :-

$$M_{prewitt} = 226.27$$

$$226.7 < 250$$

Prewitt does not detect edge

Numerical Expansion :-

sobel produces largest gradient magnitude

$$339.41 > 226.27$$

5) & Global Thresholding - iterative and Otso segmentation

intensity rk	0	1	2	3	4	5	6	7
frequency nk	3	6	4	1	1	5	8	4

Total pixel:

$$N = 32$$

(c) Gradient direction:-

$$\alpha = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

$$= \tan^{-1}\left(\frac{240}{240}\right)$$

$$\boxed{\alpha = 45^\circ}$$

(d) prewitt operator

prewitt G_x :- $G_x = (-10+10)+(-10+90)+(-10+90)$

$$\boxed{G_x = 160}$$

prewitt G_y :- $G_y = (-10-10-10)+(10+90+90)$

$$\boxed{G_y = 160}$$

Magnittude :-

$$M = \sqrt{160^2+160^2}$$

$$\boxed{M = 226.27}$$

Approximation:-

$$M = 160+160$$

$$\boxed{M = 320}$$

percentage error:-

$$\frac{320-226.27}{226.27} \times 100$$

$$= \boxed{41.42\%}$$

(a) Global Mean intensity

$$M_G = \frac{\sum r_k n_k}{N}$$

$$= \frac{0(3) + 1(6) + 2(4) + 3(1) + 4(1) + 5(5) + 6(8) + 7(4)}{32}$$

$$= \frac{122}{32}$$

$$\boxed{M_G = T_0 = 3.8125}$$

(b) iterative global thresholding

Starting threshold: $T_0 = 3.8125$

Iteration 1:

class C_1: levels 0, 1, 2, 3

class C_2: levels 4, 5, 6, 7

Mean of C_1.

$$M = \frac{17}{14}$$

$$\boxed{M_1 = 1.2143}$$

Mean of C_2

$$M_2 = \frac{105}{18}$$

$$\boxed{M_2 = 5.8333}$$

updated threshold:-

$$T_1 = \frac{m_1 + m_2}{2}$$

$$= \frac{1.2143 + 5.8333}{2}$$

$$\boxed{T_1 = 3.5238}$$

$$\Delta T = |3.5238 - 3.8125|$$

$$\boxed{\Delta T = 0.2887}$$

$$\boxed{0.2887 < 0.5}$$

$$\boxed{T = 3.524 \approx 3.5}$$

(c) Otso's method:-

formula :-

$$\sigma^2_B(k) = P_1(k) P_2(k) [m_1(k) - m_2(k)]^2$$

for $k = 2$

class $G = 0, 1, 2.$

$$P_1 = \frac{13}{32} = 0.40625$$

$$P_2 = \frac{19}{32} = 0.59375$$

$$m_1 = \frac{14}{13} = 1.0769$$

$$m_2 = \frac{108}{19} = 5.6842$$

Therefore,

$$\sigma^2_B(2) = 5.1202$$

for $K = 3$.

class $C_1 = 0, 1, 2, 3$

$$P_1 = 0.4375$$

$$P_2 = 0.5625$$

$$m_1 = 1.2143$$

$$m_2 = 5.8333$$

$$\boxed{\sigma^2_B(3) = 5.2506}$$

for $K = 4$

class $C_1 = 0, 1, 2, 3, 4.$

$$\boxed{\sigma^2_B(4) = 5.1354}$$

$$P_1 = 0.46875$$

$$P_2 = \frac{17}{32} = 0.53125$$

$$m_1 = 1.4$$

$$m_2 = 5.9412$$

Maximum variance occurs at

$$\boxed{K = 3}$$

(d) Binary Mapping and object percentage

Since the image contains dark objects on a bright conveyor belt, the lower intensity levels represent the objects

Otsu threshold $k^* = 3$

Mapping

intensity	Binary Output
0	1 - object
1	1 - object
2	1 - object
3	1 - object
4	0 - background
5	0 - background
6	0 - background
7	0 - background

object pixels : $3 + 6 + 4 + 1 = 14$

percentage : $\frac{14}{32} \times 100 = 43.75\%$

(e) Comparision of two methods :-

iterative threshold $T = 3.524$

otsu threshod $k^* = 3$

The histogram is suitable for global thresholding